

KHIN JOSMAR V. MISUARI

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I am an aspiring theoretical physicist with a research background in theoretical condensed matter physics and quantum gravity phenomenology, actively transitioning toward High Energy, Cosmology, and Astroparticle Physics (HECAP) through the ICTP Postgraduate Diploma Program.

RESEARCH INTERESTS

Gravitational Physics · Early Universe Cosmology · Quantum Gravity Phenomenology · Gravitational Waves

EDUCATION

The Abdus Salam International Centre for Theoretical Physics (ICTP) Trieste, Italy
Incoming Postgraduate Diploma in High Energy, Cosmology, and Astroparticle Physics Sept. 2026 –

- One of 10 students selected globally for the HECAP field; 5th Filipino admitted to HECAP since the program's inception and the first from the University of San Carlos and Cebu province.

University of San Carlos Cebu City, Cebu, Philippines
Bachelor of Science in Applied Physics Aug. 2019 – May 2023

- **Cumulative GPA: 1.42 / 1.00** (Note: 1.00 = Highest/Excellent; 3.00 = Minimum Pass)
- **Graduate-Level Coursework** (non-credit, audited):
 - **General Relativity** · Grade: **1.00 / 1.00**, Ranked **1st** among M.Sc. & PhD candidates · SY 2021–2022, 2nd Semester
 - **Mathematical Methods in Physics** · Grade: **1.00 / 1.00**, Ranked **1st** among M.Sc. & PhD candidates · SY 2021–2022, Summer Term
- **Theoretical Electives:**
 - **Classical Mechanics II** (PHYSEL 4105) · Grade: **1.00 / 1.00**
 - **Quantum Mechanics II** (PHYSEL 4206) · Grade: **1.00 / 1.00**

Carcar City Alternative Learning System Carcar City, Cebu, Philippines
High School Diploma via Alternative Learning System (ALS) Mar. 2019 – Jun. 2019

- Bypassed four years of traditional high school through the national Accreditation & Equivalency (A&E) Test, advancing directly from Grade 8 to university.

RESEARCH EXPERIENCE

OJT Research Project Jun. 2022 – Jul. 2022
University of San Carlos · Supervisor: Dr. Roland Emerito S. Otadoy Cebu City, Philippines

- Independently reproduced results of a peer-reviewed publication in **quantum gravity phenomenology**: Abac, Esguerra, & Otadoy, *Int. J. Mod. Phys. D* **30** (2021) 2150005. Derived GUP-modified stellar structure equations and numerically reproduced mass-radius relations for zero-temperature white dwarfs.

Undergraduate Thesis · 2nd Place Best Thesis Award Aug. 2022 – May 2023
University of San Carlos · Adviser: Dr. Danilo M. Yanga Cebu City, Philippines

- Title: *Calculation of the Eliashberg Function of Particles in Graphene in the Migdal-Eliashberg Theory of Superconductivity*
- Derived purely analytic expressions for the Eliashberg function and phonon linewidth in doped graphene using many-body formalism. Awarded **2nd Place** Best Thesis by the Department of Physics.

Initiator & Program Head, ASCEND: Physics

Jul. 2023 – May 2026

Sacred Heart School – Ateneo de Cebu

Mandaue City, Cebu, Philippines

- Designed and implemented an initiative to cultivate a culture of physics excellence in the province, designing an original university-level enrichment program to establish a rigorous theoretical foundation where none previously existed.
- Transformed the school's competitive standing from zero podium or notable placements in any national physics or astronomy competitions to the following selected outcomes:
 - ◊ **Philippine Physics Olympiad (PPO)**
 - **PPO 2026: 4 out of 32 National Finalists (2nd highest nationwide**, behind only Philippine Science High School Main Campus). **7 Qualifying Round Merit Awardees (1st in the Visayas, 3rd nationally per school).**
 - **PPO 2025:** 2 out of 27 National Finalists (first in school history)
 - ◊ **University-hosted Competitions**
 - **UP Big Bang Astronomy Quiz Show 2026: National 1st Runner-Up.** Outperformed Philippine delegates of the International Olympiad on Astronomy and Astrophysics (IOAA) from Philippine Science High School.
 - Other notable finishes: **UP PISIKAAalamanan 2026 1st Runner-up, Ateneo Physics Olympiad 2024 2nd Runner-up, USC Astronomy Quiz Bowl Champion (2023, 2024, 2025).**

Member, Carolinian Physics & Astronomy Society

Aug. 2019 – May 2023

University of San Carlos

Cebu City, Cebu, Philippines

- **Chairman of Academic Affairs, SY 2021–2022** · Led several structural departmental reforms such as inclusion of Classical Mechanics II as an elective in the B.Sc. Applied Physics program and structural revisions to the Mathematical Physics syllabus, re-sequencing advanced topics to better align with international standards for theoretical research.
- **Core Member, USC Quiz Bowl Series Question Committee** · Authored rigorous, university-level problems for the annual USC Physics and Astronomy Quiz Bowl inter-school competition.
- Acted as an informal peer mentor for undergraduate physics majors, providing academic support and facilitating review sessions on topics ranging from introductory physics and calculus to advanced subjects like General Relativity and Mathematical Physics.

TEACHING EXPERIENCE

Senior High School Faculty & Olympiad Head Coach

Jul. 2023 – May. 2026

Sacred Heart School – Ateneo de Cebu

Mandaue City, Cebu, Philippines

- **DOST-JLSS Return Service Obligation.** Fulfilled mandatory scholarship service as a physics instructor and pioneered academic competition culture in the school.

AP/IB Physics & Mathematics Tutor

Jun. 2023 – Present

Freelance

Cebu, Philippines

- Provides specialized instruction in high school and introductory university mathematics and physics.

Teaching Assistant (TA)

Jan. 2023 – May 2023

University of San Carlos, Department of Physics

Cebu City, Philippines

- Selected by physics faculty to assist a large-class of 100+ engineering students for *Physics for Engineers*. Evaluated student assessments and conducted regular office hours to provide targeted academic support.

RELEVANT WORKSHOPS & TRAINING

1st Jagna Space Encounters | Central Visayan Institute Foundation, Bohol

Jan. 2025

- Intensive workshop on astrophysics and space science. Developed practical skills in Python for astronomical data analysis and computational modelling.

AWARDS & HONORS

2020 – 2023 Dean's Honors List · **Ranked 10th (2020), 1st (2021), 2nd (2022)** · University of San Carlos

2023 2nd Place Best Thesis Award · Department of Physics, University of San Carlos

2022 Iron Wood Property Ventures Corporation Merit-Based Undergraduate Scholarship

2021 DOST Junior Level Science Scholarship (DOST-JLSS) · Granted to top STEM undergraduates in the Philippines

2019 ALS National Top Achiever · Exceptional score on A&E Test; advanced from Grade 8 directly to university

TECHNICAL SKILLS

Programming: Python, R, MATLAB

Data Analysis: pandas, NumPy, SciPy, dplyr, Excel

Data Visualisation: Matplotlib, ggplot2

Developer Tools: VS Code, PyCharm, Jupyter Notebook

Scientific Writing: L^AT_EX, technical writing, experiment design

Languages: Cebuano (native), Filipino (fluent), English (fluent)